Joint host-pathogen genomic analysis identifies hepatitis B virus mutations associated with human NTCP and HLA class I variation

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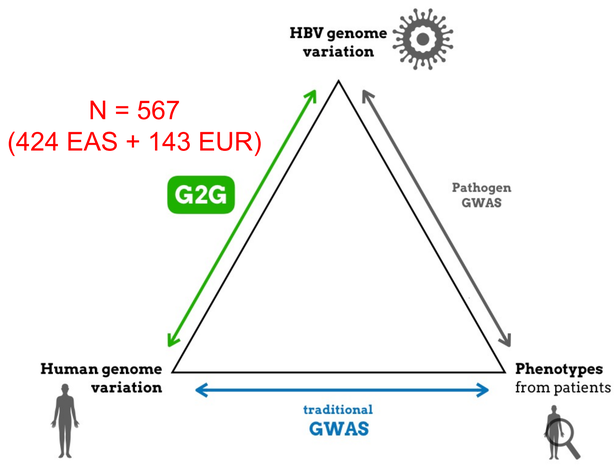
INTRODUCTION

- Host-pathogen co-evolutionary "arms race" shapes both host and pathogen genetic diversity.
- Paired host and pathogen sequences to pinpoint sites of co-evolution for hepatitis B virus (HBV).
- Searched for associations between host and pathogen variation under a genome-to-genome (G2G) framework
- Identified HBV escape mutations induced by host selective pressure originating from:
    1. NTCP S267F
    2. HLA-A variation

METHODS

Genome-to-genome (G2G) Framework

Hepatitis B virus (HBV):
- Ultra-deep sequencing
- Baseline (before antiviral therapy)



Human:
- Exome sequencing
- Chronic infection

A GWAS for each non-synonymous HBV mutation:

$$y_j \sim \boldsymbol{\beta_{ij}} G_i + \sum_k \alpha_k X_k$$

j^{th} HBV Mutation (Absence or Presence)

i^{th} Host Variant

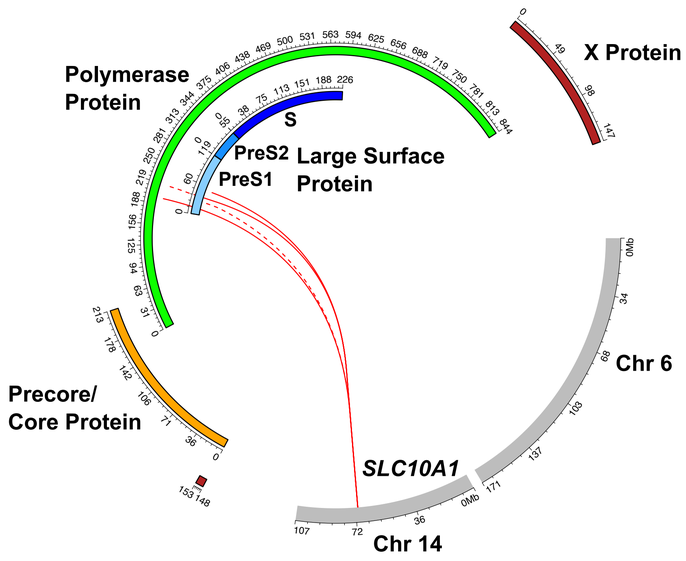
Covariates: Age, sex, host and pathogen PCs

Correction for host and viral stratification
- False positives due to shared geography

True associations -> HBV mutations that are advantageous under certain host genetic backgrounds:
1. Differences in susceptibility or virulence
2. Intra-host evolution

RESULTS

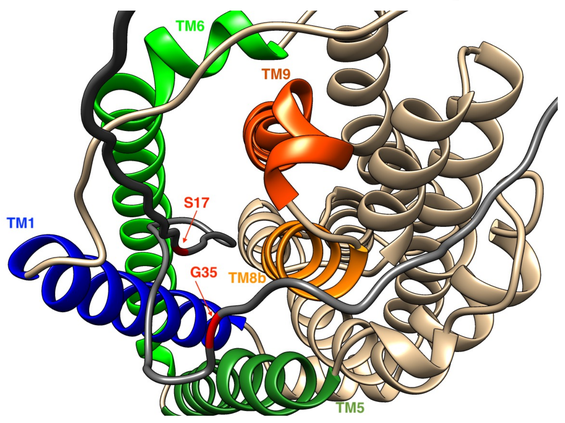
1. NTCP S267F and HBV preS1 mutations

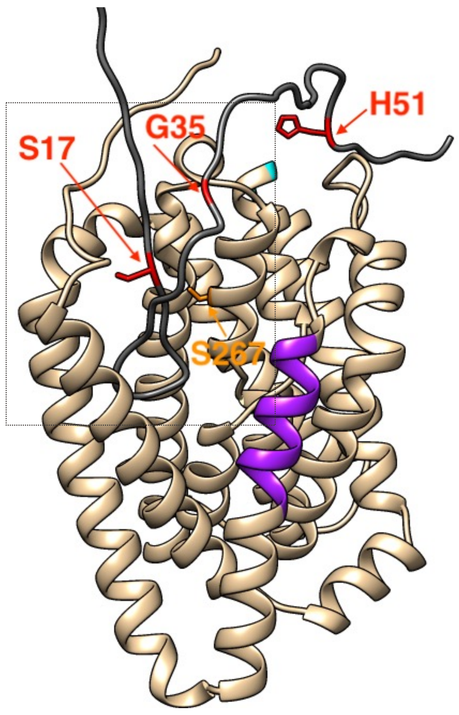


rs229651 is an East-Asian specific missense SNP in *SLC10A1* (NTCP S267F)

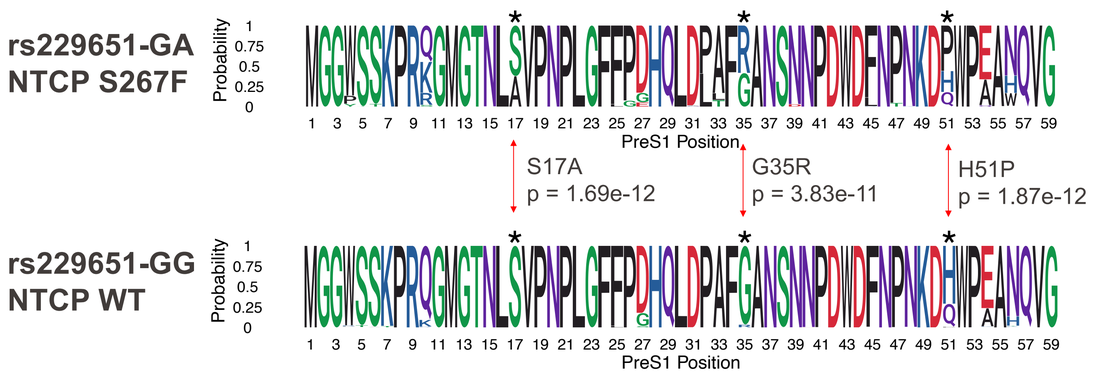
NTCP S267F inhibits viral entry -> associated with favorable Hepatitis B prognosis

HBV preS1 mutations are located within sites relevant for preS1-NTCP binding:

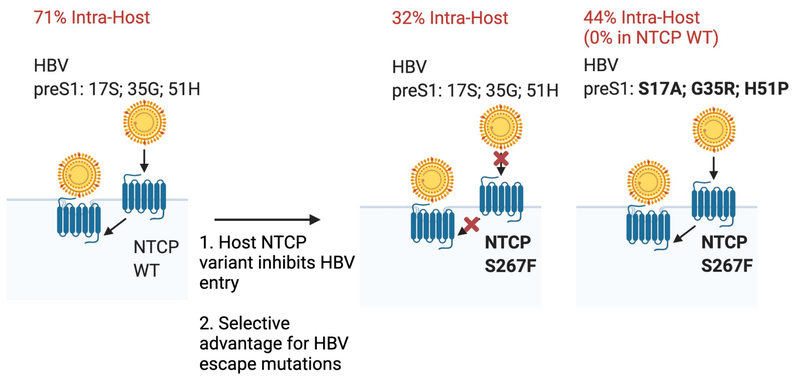




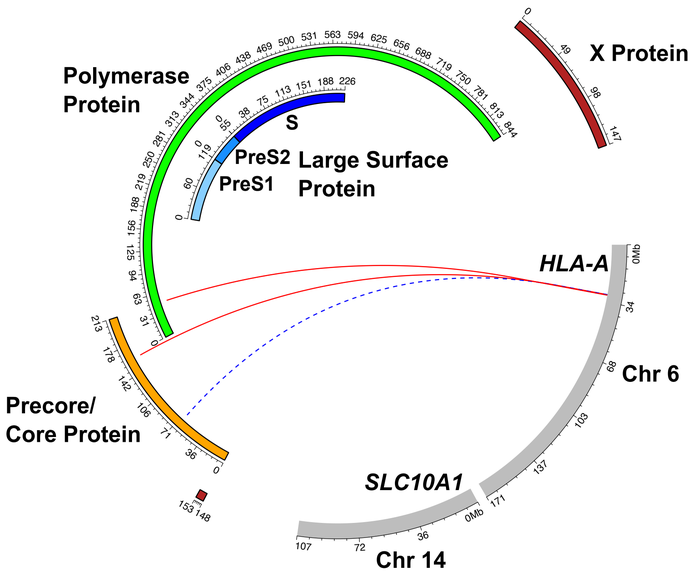
NTCP S267F is associated with three HBV preS1 residues:



Hypothesis: HBV escape mutations counteract inhibitory effect of NTCP S267F on viral entry:



2. HLA-A variation and HBV epitope mutations



East Asian

1) HLA-A*02:06 ↔ VNIPWTHKV (P = 1.3e-12, OR = 84.2)
Binding Affinity (PR): 0.49%
Pol; Pos 49

2) HLA-A*33:03 ↔ WIRTPPAYR (P = 2.1e-10, OR = 0.08)
Binding Affinity (PR): 0.05%
PreCore/Core; Pos 160

European

3) HLA-A*01:01 ↔ LLDTASALY (P = 9.7e-8, OR = 0.07)
Binding Affinity (PR): 0.01%
PreCore/Core; Pos 67

Mutations that decrease binding affinity are favored, suggesting immune evasion:

